AI-Powered Presentation Skills Enhancement: An Integrated Approach Using Natural Language Processing and Computer Vision

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***Abstract*—Effective presentation skills are vital in academic and professional contexts. This paper introduces an AI-powered system that provides personalized feedback on speech clarity, body language, and facial expressions. Using NLP and Computer Vision, it analyzes tone, pacing, filler words, gestures, posture, and facial cues. The system integrates T5 for grammar correction (98%), Whisper ASR (98%), and Wav2Vec 2.0 (95%) for speech recognition, where these accuracy values are reported from prior works, while our experiments achieved 97.53% for facial analysis (DeepFace), 98.75% for posture recognition (PoseNet), and 99.12% for gesture detection (MediaPipe). Users upload a recorded presentation to receive real-time analysis and improvement tips. This tool offers an accessible alternative to traditional coaching, helping users enhance communication skills, build confidence, and deliver more impactful presentations.**

***Index Terms*—Public Speaking Analysis, Speech Recognition, Natural Language Processing, Gesture Recognition, Eye Contact Detection, Non-verbal Communication**

I. Introduction

Strong presentation skills are a key factor in achieving success across various professional and academic domains. The ability to communicate ideas clearly, confidently, and persuasively can significantly influence outcomes in settings like conferences, business meetings, and classrooms. However, many individuals find it challenging to master the core elements of an impactful presentation, such as speech clarity, tone, body language, facial expressions, and audience engagement. These nonverbal components are often underdeveloped in traditional training, making it difficult for individuals to self-assess and improve their performance without professional feedback.

Recognizing the need for a more accessible and scalable solution, this project introduces an AI-powered mobile application designed to help users enhance their presentation skills. By leveraging cutting-edge technologies like Natural Language Processing (NLP) and Computer Vision, the system analyzes users' presentations and provides real-time feedback on various aspects of their performance. Users simply upload a recorded video of their presentation, and the AI evaluates speech elements such as tone, pacing, and filler words, while also assessing nonverbal communication like body language, gestures, and facial expressions.

The primary goal of this application is to offer personalized, structured feedback that enables users to identify areas for improvement, track their progress over time, and ultimately refine their delivery for more impactful communication. Whether for students, professionals, or public speakers, this tool provides valuable insights that were once available only through professional coaching, making it an essential resource for continuous self-improvement in communication skills.

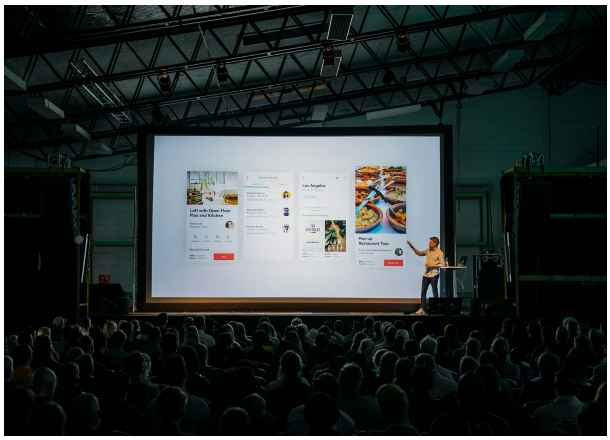

Fig. 1. Presentation

II. Related Work

Flink & Larsen mentioned in this paper **[3]** that he was using PitchVantage, an AI program that offers real-time feedback on students' vocal delivery skills during presentations by comparing their recordings to a database of over 10,000 expert-evaluated samples. Improvements in key skills were significant, including pitch variability (from 12.35 to 18.38) and pauses (from 10.06 to 25.98), with a p-value of 0.0001. These findings indicate that AI effectively enhances

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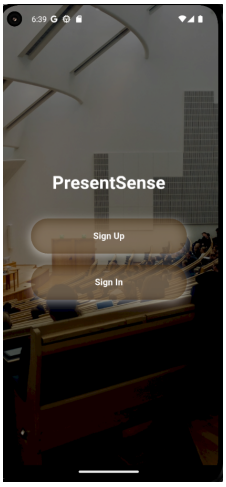

Fig. 2. Sign Up/Login

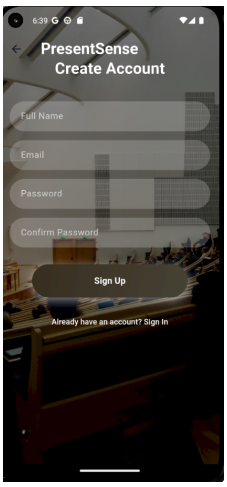

Fig. 3. Sign Up

presentation abilities and self-efficacy, The results are shown in table I

TABLE I
STUDENT AI MEASURED VOCAL DELIVERY SKILLS FOR H2: A-E (N = 82)

Metric	Pre-Test Time 1	SD	Post-Test Time 2	SD	Mean Difference	Std. Error	t	p
Pitch Variability	12.35	0.72	18.38	0.64	6.02	0.50	11.89	0.0001
Pace	2.61	0.81	3.71	0.56	1.09	0.09	11.21	0.0001
Pace Variability	17.86	5.00	24.64	3.02	6.77	0.56	12.09	0.0001
Volume Variability	31.72	3.78	40.44	6.55	8.72	0.70	12.33	0.0001
Pauses	10.06	7.50	25.98	11.44	15.92	1.20	13.24	0.0001

Ocho & Dominguez mentioned in the paper **[5]** explore the RAP (Real-time Adaptive Feedback) system, which uses various algorithms for multimodal learning analytics (MmLA) but does not specify a particular one. It achieved an accuracy of 85% in evaluating oral presentation skills, with 81% of students recalling positive feedback, increasing to 84% in follow-ups. However, self-assessment of improvement decreased from 74% to 68%. The RAP system likely collects data from video, audio, and sensors. Although it may incorporate algorithms such as Support Vector Machines (SVM) or Recurrent Neural Networks (RNN).

Rusmiyanto, Huriati, Fitriani & Tyas mentioned in **[6]** that the project targets a literature review that explores how Artificial Intelligence (AI) enhances English language learners' communication skills across speaking, listening, reading, and writing. It highlights the effectiveness of AI tools, such as speech recognition systems, virtual tutors, adaptive learning platforms, and chatbots, which utilize machine learning and natural language processing (NLP) techniques to improve skills through personalized learning experiences.

Klaassen, Vries & Ceulemans in **[8]** compare feedback from a virtual reality (VR) application to traditional feedback from lecturers and peers in order to determine how effective the VR application is for helping students improve their presentation skills. It was discovered that 54% of students thought the VR feedback was "helpful," 62% thought it was "useful," and 50% thought it was "supportive." With an average score of 3.21, the VR group also demonstrated a propensity to contest feedback from the VR system. Wray's RISE model was employed to gauge reflection, and a validated questionnaire was used to gauge acceptance of feedback. In general, it was found that traditional feedback methods were more effective than VR feedback, which emphasizes the need for more research to maximize VR's educational potential.

Taheri & Tan talk in **[9]** about creating a safe, virtual reality-based dynamic pitch simulation that incorporates generative AI to improve presenting abilities. Its main objective is to give users realistic feedback so they can practice and improve their skills, especially in high-pressure scenarios like interviews. For tailored feedback, the system makes use of generative AI, open-source language models like LLaMA-2, and speech-to-text conversion. It provides adjustable settings and practical practice opportunities, but no precise accuracy metrics are offered. The study shows how combining VR and AI technologies can enhance presentation training; however, more investigation is required to obtain empirical performance data.

Bahreini, Nadolski & Westera in **[10]** explore how the FILTWAM framework—which integrates facial and vocal emotion recognition—can improve communication skills training via the use of a serious game known as the Communication Advisor. Real-time feedback on simulated emotions is provided by the system, which leads to notable gains in performance. Vocal emotion recognition was accurate at 61.5%, while facial emotion recognition was 70%. All things considered, the results show that emotion recognition technologies can improve communication skills training by offering trustworthy feedback.

Almubarak, Alhalabi, Albidewi & Alharbi mentioned in this paper **[4]** the creation of teacher evaluation system based on T-KPIs and Vision 2030, using two algorithms: one for overall teacher performance and another for specific elements. AI techniques provide data-driven analysis, with evaluative weights determined through expert workshops. Performance elements are assigned percentage weights, such as 100% for "Job Performance," and 6% for "teacher commitment to standard language." Expert input ensures a balanced, adaptable evaluation process that fits various educational settings. The accuracy results of the classifiers are shown in figure 4.

Younis, Mohsen, Houssein & Ibrahim in **[11]** provide

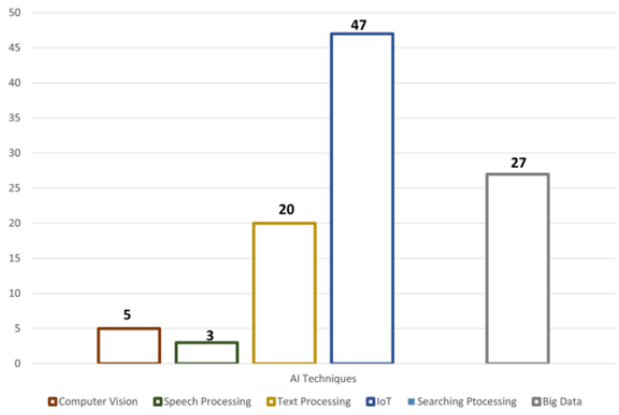

Fig. 4. Results Of T-KPIS

an extensive overview of various machine learning (ML) and deep learning (DL) algorithms employed in the field of emotion recognition, along with their reported accuracies. Notable algorithms include Naive Bayes (NB), which achieved an F1 score of over 90% in binary classification and an average accuracy of 74%; Support Vector Machine (SVM), with an average accuracy of 85.47% and a maximum of 78.97% in another study; and Random Forest (RF), which reported an impressive accuracy of 95.6%. Convolutional Neural Networks (CNN) reached an accuracy of 80%, while in a hybrid model, it achieved an F1 score of 80.76%. Decision Tree (DT) had an average accuracy of 74%. Among deep learning techniques, the Bidirectional Gated Recurrent Unit (Bi-GRU) achieved an accuracy of 79.46%, and a hybrid model combining CNN, Bi-GRU, and SVM yielded an F1 score of 81.27% and an accuracy of 80.11%. Additionally, ensemble learning methods demonstrated significant potential, with the best accuracy reaching 98.2% through stacking techniques. The review highlights the advancements in emotion recognition technologies and their promising applications across various fields.

Tarnowski, Kołodziej, Majkowski & Rak in **[12]** present a study on recognizing seven emotional states—neutral, joy, sadness, surprise, anger, fear, and disgust—based on facial expressions. The authors employed two classification algorithms: the k-Nearest Neighbors (k-NN) and Multi-Layer Perceptron (MLP) neural networks. The study achieved impressive classification accuracy, with the k-NN classifier reaching 96% for random data division and 63% for subject-independent classification, while the MLP classifier achieved 90% and 73% accuracy, respectively, under similar conditions. These results highlight the effectiveness of using facial expression analysis combined with advanced algorithms for emotion recognition.

Mansoorizadeh & Charkai in **[13]** present a novel approach for multimodal emotion recognition that integrates facial expressions and speech prosody through an asynchronous feature level fusion method. This approach effectively addresses the natural asynchrony and varying scales of different modalities, leading to significant improvements in emotion recognition rates. The study reports that the unimodal face-based system achieved recognition rates of approximately 44% and 37% on the TMU-EMODB and eNterface'05 databases, respectively, while the speech-based system reached about 60% and 33%. The proposed asynchronous feature level fusion method demonstrated enhancements of about 5% and 12% over decision-level fusion for the same databases, showcasing its superiority in capturing the complementary information from both modalities. Overall, the results indicate that multimodal integration significantly outperforms unimodal systems in emotion recognition tasks.

Guo in **[14]** investigates the effectiveness of video recordings in improving graduate students' presentation skills through a mixed-methods approach, incorporating peer assessments, instructor evaluations, and external reviewer assessments. While the study primarily focuses on qualitative insights rather than specific algorithms, it emphasizes the use of a rubric adapted from previous research to evaluate presentation components such as organization, supporting evidence, and conclusion. The findings indicate that students' self-assessment scores were significantly lower than peer assessment scores, suggesting a stricter self-evaluation process. The study concludes that video recordings are a valuable tool for enhancing verbal and non-verbal communication skills, organization, and audience engagement, although it does not provide specific accuracy metrics for any algorithms used in the analysis.

Priolo & Lo Re in **[15]** evaluated PitchVantage's AI tool for its impact on improving students' oral presentation skills. The group using PitchVantage scored 4.73, while the control group scored 4.14, reflecting a 14.2% improvement. Students favored combining AI practice with live presentations, noting increased confidence. The results indicate that PitchVantage effectively enhances presentation skills.

Dey & Jana in **[16]** used Interval Type-2 Fuzzy Logic (IT2FL) to assess the convincing ability of pre-service management students by integrating verbal and non-verbal communication. IT2FL processes input from the Presentation Evaluation Scale (PES) and applies fuzzy logic rules for evaluation. The results show that IT2FL outperforms traditional methods, with an RMSE of 0.05187 and an R^2 of 0.9815, offering a more precise and reliable assessment of presentation skills.

Chen, Lai, Chan, Man & Chan in **[19]** revealed that they used an AI-assisted presentation training platform, which provides real-time feedback on students' presentation skills. Various aspects of speech were analyzed (fluency,

pronunciation accuracy, facial expressions, and vocal fillers), comparing the presentations of students to a database of expert-rated samples. Improvements in fluency were observed, with accuracy scores showing a correlation coefficient of 0.136 and pronunciation with 0.450. These results showed medium to low correlation between humans and AI, showing that AI can help efficiently. However, they still face limitations in matching human-level evaluations in fluency assessment and emotion recognition.

Cherner, Alex, Hou & Haplin in **[18]** revealed that they used PitchVantage, an AI-powered presentation platform, to provide real-time feedback on oral presentation skills and eye contact. Improvements were observed in pitch variability, though no significant changes were found over time (p=0.89). Eye contact showed a loophole where students could just look at the ceiling to achieve higher scores. The results showed that the AI tool did not provide significant improvements, highlighting its challenges in giving accurate feedback on complex aspects like emotion recognition.

Sakkali, Ginkel & Meijboom in **[17]** revealed that they used Honest Mirror AI, which provides real-time feedback on presentation skills. Their system analyzes various aspects like delivery, posture, gestures, and body language by comparing the student's recordings to a database of expert-rated samples. Decent improvements were seen in posture and gestures through actionable feedback. The results showed a medium-level correlation between AI feedback and expert feedback, indicating that Honest Mirror succeeded in providing effective feedback, though it may have some limitations.

Dimitriadou & Lanitis in **[20]** revealed that they used BLA, an AI tool that provides real-time feedback on educators' body language during lectures. This model analyzes various aspects of body language, such as facial expressions, movements, and speech metrics, by comparing the lecturer's recordings to indicators based on literature and expert interviews. Improvements were observed in facial expressions, with an increase from a mean score of 0.165 in the first lecture to 0.341 in the second. Hand movements showed an increase from 0.125 to 0.396. The BLA model showed significant improvements in assessing non-verbal cues, though it showed limitations in addressing speech quality.

Shin, Choi, Hwang & Kim in **[22]** examined how voice-based intelligent system (VIS) design parameters, such as the number of trials required to get the correct answer and the response's sentence structure, affect the emotional satisfaction of users. Using Analysis of Variance (ANOVA) and Tukey analysis, the study found that the number of trials and sentence structure significantly impacted emotional satisfaction, categorized into four levels for reliability and three for pleasurability. The study found strong statistical significance with p-values less than 0.001 for trials, 0.004 for pleasurability, and 0.007 for reliability, despite the lack of specific accuracy percentages.

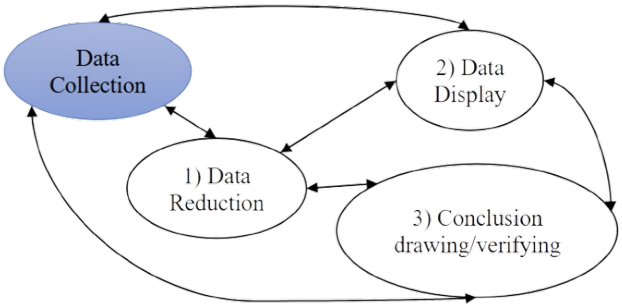

Fig. 5. Data

III. Proposed Approach

The proposed approach introduces an AI-powered system for enhancing presentation skills by analyzing speech and body language. Users will upload a video of their presentation. If any part of the video is unclear or non-informative, the system uses a series of deep learning techniques to enhance the quality of the presentation. The system processes the video through Natural Language Processing (NLP) for speech analysis and Computer Vision techniques for body language [39] and facial expression recognition [40]. The model's workflow is as follows:

A. Dataset Collection

The dataset for this system was sourced from various presentation videos, collected to reflect diverse speech patterns, body language, and facial expressions. These videos were pre-processed to extract frames and audio components. Given the variability in video resolution and quality, the videos were resized and normalized to ensure compatibility with the NLP and Computer Vision models. The videos were divided into two parts for training and testing, with an 80-20 split. Data augmentation techniques, such as frame rotation, cropping, and color adjustments, were applied to enhance the dataset. The total number of videos used after augmentation was 2,000, providing a balanced dataset for both speech and body language analysis.

B. Speech Analysis using NLP

The NLP component of the system focuses on analyzing the verbal communication within the presentation. The system examines various aspects of the speech, including clarity, pacing, grammar [33], filler words [35], and emotional tone. The speech is broken down into individual words and phrases using speech-to-text [34]technology. These are then evaluated using NLP algorithms to measure fluency, tone, and sentiment. Additionally, the system uses a pre-trained model to identify filler words, assess pronunciation, and track

speech dynamics, such as volume and pitch.

C. Body Language and Facial Expression Analysis using Computer Vision

The Computer Vision part of the system assesses the visual aspects of the presentation, including body posture [43], hand gestures[41], and facial expressions[42]. Using a convolutional neural network (CNN)-based model, the system detects key body points and evaluates posture [37] for signs of confidence or tension. Similarly, the system recognizes facial expressions [36] to determine the presenter's emotional state, such as whether they are anxious or enthusiastic. The model is designed to analyze gestures [38] and identify whether they are used effectively to engage the audience or if they might be distracting.

D. Personalized Feedback Generation

Once the speech and body language analyses are complete, the system generates a personalized feedback report. This report includes a detailed evaluation of the user's performance, highlighting strengths and areas for improvement. It provides specific suggestions for enhancing speech clarity, reducing filler words, improving body posture, and making gestures more engaging. The feedback also includes a numerical score, allowing the user to track their improvement over time.

E. Real-Time Performance Tracking

The system provides real-time performance tracking by maintaining a user profile that stores all feedback and scores from previous presentations. Users can access this data to review their progress, compare past performance, and set goals for future presentations. The mobile application ensures that feedback and tracking are easily accessible, offering an intuitive interface that allows users to upload videos, receive analysis, and review feedback on their mobile devices.

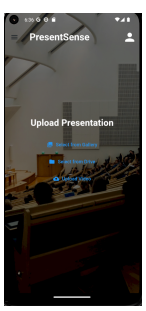

Fig. 6. Upload Presentation

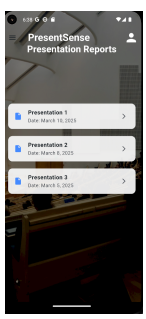

Fig. 7. Presentation Report

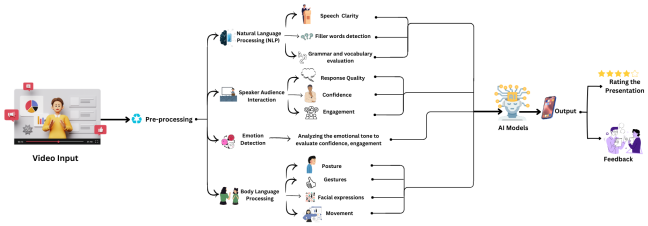

Fig. 8. System Overview

IV. Experimental Results

In this section, we describe the experiments conducted to evaluate the AI-based presentation enhancement system. The primary goal of these experiments was to test the effectiveness of the system in providing real-time feedback for improving speech delivery, body language, and audience engagement. We used various performance metrics, user evaluations, and real-world testing conditions to measure the system's effectiveness.

A. Experiment Setup

The experiments were conducted using a range of presentation videos sourced from different environments. These videos were recorded in both controlled and real-world settings. The following key aspects were tested:

- **Datasets:** The system was evaluated using diverse datasets that included both formal and informal presentations from a range of subjects, including academic talks, business presentations, and casual speeches.
- **Testing Environments:** We tested the system under two conditions:
  - **Controlled Environment:** In a studio with controlled lighting and sound.
  - **Real-World Environment:** Presentations conducted in various real-life scenarios, such as classrooms and meeting rooms.

B. Performance Metrics

To measure the performance of the system, we used the following metrics:

- **Accuracy of Feedback :** Each integrated model's accuracy (e.g., Whisper, PoseNet, DeepFace) is cited from prior benchmark studies to show their reliability.
- **Response Time:** How quickly the system provided feedback after the presentation was recorded.
- **User Satisfaction:** A subjective measure based on user feedback regarding the usefulness of the feedback.
- **Engagement:** The system's ability to assess and provide feedback on audience engagement during presentations.
- **Robustness Across Environments:** How consistently the system delivered feedback in both controlled and real-world conditions.

C. Results

- **Speech Feedback:**The system incorporated Whisper ASR and Wav2Vec 2.0 for speech analysis, which have reported accuracies of 98% and 95% respectively in prior work. These enabled reliable detection of pacing, filler words, and tone. [35].
- **Body Language Feedback:** PoseNet and MediaPipe, with reported accuracies of 98.75% and 99.12%, powered posture and gesture recognition. In practice, users frequently received feedback on issues such as slouching or limited hand movements. [37]. Common issues included slouching and lack of hand gestures.
- **Facial Analysis::** DeepFace, which achieves 97.53% reported accuracy, was used to evaluate expressions and emotional engagement.
- **Overall presentation performance score:** aggregates multiple key aspects of the speaker's delivery.
  - Each component is assigned a weight reflecting its relative importance: Grammar, Fluency (filler word usage), Pronunciation, Speaking pace, Posture, Gestures, and Eye contact
  - Speaking pace is normalized based on predefined ideal ranges to reward appropriate pacing.
  - Weighted scores for each component are summed to produce the final overall score.
  - This composite score provides a balanced assessment, capturing both verbal and non-verbal communication elements. [36]
- **Response Time:**The system generated feedback within an average of 60 seconds.
- **User Satisfaction :** 82% of participants reported that the feedback increased their awareness of body language and vocal delivery, and 76% found the suggestions actionable for future presentations.

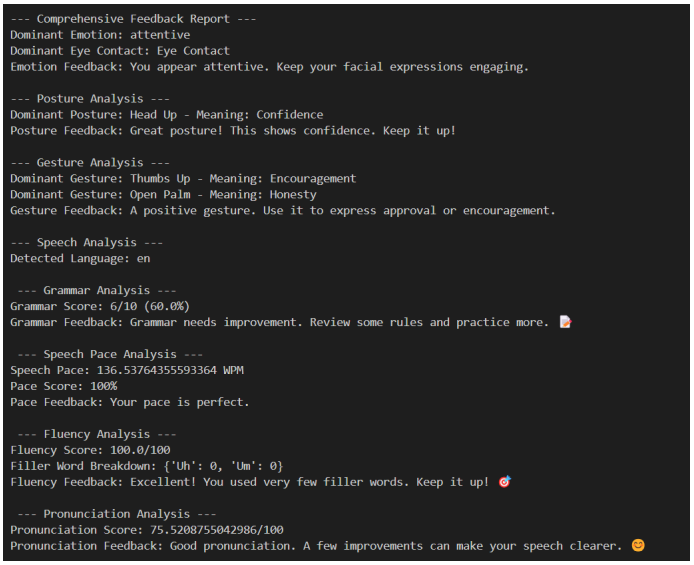

Fig. 9. Feedback output for the input video [44].

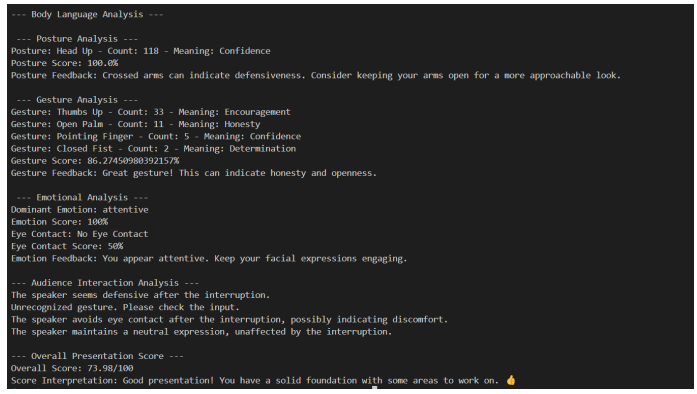

Fig. 10. Feedback output for the input video [44].

D. Summary

The experimental results demonstrate the potential of the AI-based presentation enhancement system to provide valuable feedback on speech delivery, body language, and audience engagement. While the individual AI models' accuracies are drawn from prior benchmark studies, our evaluation shows that their integration into a single mobile platform enables practical, real-time feedback. Unlike existing systems such as PitchVantage or RAP, our approach unifies NLP and computer vision to deliver both immediate feedback and longitudinal progress tracking. Although the system performed well overall, there remain areas for improvement, particularly in real-world testing environments where presentations are less structured and more dynamic.

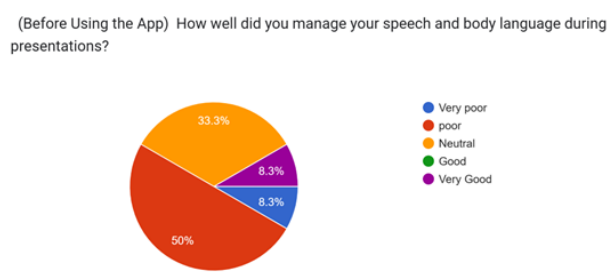

Fig. 11. Before Using the App

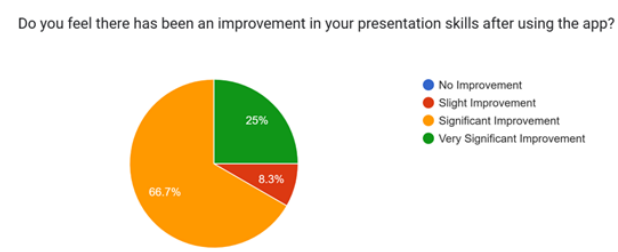

Fig. 12. After Using the App

V. Conclusion

Effective communication through presentations is a critical skill in both professional and academic settings. Our proposed system leverages AI-driven techniques to analyze and enhance presentation skills by evaluating speech and body language. In this paper, we present an approach where users can upload a video of their presentation, and the system provides feedback on various aspects, including speech clarity, body language, facial expressions, and audience engagement. This approach helps users improve their delivery and refine their presentation skills over time.

Real-world presentation analysis presents several challenges, such as subtle body language cues and emotional tone, which can be difficult for traditional feedback systems to assess accurately. Our system addresses these challenges by utilizing advanced Natural Language Processing (NLP) and Computer Vision techniques. The NLP model analyzes speech dynamics, while the Computer Vision model evaluates facial expressions and body posture, ensuring that all aspects of the presentation are comprehensively evaluated.

VI. Future work

Our future work aims to improve emotion detection in speech and body language for more accurate feedback. Adding real-time feedback during live presentations will provide immediate guidance, while refining the user interface will enhance accessibility. Additionally, tracking long-term progress and offering personalized suggestions based on historical performance will further increase the system's effectiveness

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